Evaluation Metrics and Rewards for AI-Based Telescope Schedulers

A validated effective-exposure-time metric and per-night schedule scores on real DES data (v0.1)

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Abstract

Before we can train or even fairly compare AI schedulers for ground-based telescopes, we need a number that says how good an observation—and a night of observations—actually was. This note documents the first phase of that work. We adopt the *effective exposure time* $t_{\rm eff}$ of Neilsen et al. (2019), the same quantity used as the reinforcement-learning reward by Terranova et al. (2023), and reconstruct it from first principles directly from the quality columns of the public DES exposure log (105,889 exposures, 2013–2019). A per-band calibration of the fiducial seeing and of an empirical sky-brightness term brings our reconstruction to a log-correlation of 0.95 against the DES pipeline's own $t_{\rm eff}$, with a median relative error of 15% (versus 24% for uncalibrated paper constants); the fitted fiducials independently recover the atmospheric $\lambda^{-0.2}$ law. Decomposing the blur term, we find a striking selection effect: pooled over all exposures, image blur apparently *improves* with airmass (fitted exponent -0.23 instead of the physical $+0.6$), because the DES scheduler preferentially pointed far from zenith on nights of good seeing. Within-night fits recover the physical trend. This is Simpson's paradox in scheduler logs, and it is direct evidence that evaluating scheduling policies on logged data is confounded by the logging policy—precisely the off-policy problem that motivates conservative offline RL. Finally, we aggregate $t_{\rm eff}$ into per-night schedule metrics and a transparent composite score, and apply them to 615 real DES nights. All code is packaged as the `skai_metrics` toolkit with four automated validation tests, all passing.

1 Introduction and motivation

Scheduling a ground-based telescope is a sequential decision problem of unpleasant size. At every moment the scheduler must choose a pointing, a filter, and an exposure time from a combinatorially large set of options, under observing conditions (seeing, clouds, sky brightness) that evolve stochastically on timescales of minutes, and under scientific constraints (cadence requirements, footprint coverage, airmass limits) that couple decisions made hours or nights apart. The underlying optimization is NP-hard, and production schedulers are, in practice, carefully hand-tuned heuristics.

Our group's research direction is to learn scheduling policies from data. Because a telescope cannot be handed over to an untrained agent for millions of exploratory episodes, the natural setting is *offline* reinforcement learning: train on the logs of an existing scheduler, without further interaction with the environment. Terranova et al. (2023) demonstrated the feasibility of this program by training a deep Q-network to schedule the Stone Edge Observatory, using the effective exposure time $t_{\rm eff}$ as the reward signal; on the algorithmic side, Conservative Q-Learning (CQL;

Kumar et al., 2020) is the offline-RL method we intend to build on, since it is designed to behave sensibly exactly where the logged data is silent.

Everything in this program—the reward the agent maximizes, the evaluation that decides whether the learned policy beats the baseline, the diagnostics that tell us *why*—rests on one foundation: a quantitative, physically meaningful, validated metric for how good an observation is. If the metric is wrong, the agent will faithfully optimize the wrong thing. This document describes our first contribution: the choice, physical derivation, empirical calibration, and validation of that metric on real Dark Energy Survey (DES) data, and its aggregation into schedule-level scores. We deliberately work through the physics from first principles; part of the goal of this phase was for us to *understand* the reward we are about to hand to a learning agent, not merely to import it.

2 Data: the DES exposure log

We work with `des-exposures.csv`, a public log of 105,889 exposures taken by the Dark Energy Camera (DECam) at the Blanco 4-m telescope, CTIO, between 2013-08-31 and 2019-01-10—essentially the full six-year Dark Energy Survey. Each row is one exposure, with the columns most relevant to us:

- `qc_fwhm` — delivered image full width at half maximum (arcsec), as measured by the DES quality pipeline: the "blur" of the exposure.
- `qc_cloud` — atmospheric extinction attributable to clouds. As we show in Section 4, this column is an extinction in *magnitudes* (0 = photometric).
- `qc_sky` — sky-brightness excess over nominal dark sky. Empirically this behaves as a *fractional* excess (0 = nominal dark sky; a bright moon can push it to tens).
- `teff` (and `qc_teff`) — the DES pipeline's own effective exposure time, our ground truth for validation.
- `airmass` — the secant of the zenith angle at which the exposure was taken: the path length through the atmosphere relative to pointing straight up.
- `filter` — the DECam band (g, r, i, z, Y, from blue to near-infrared).
- `program` — which observing program the exposure served (`survey`, supernova fields, GW follow-up, engineering).

For calibration and per-night scoring we restrict to the homogeneous core of the survey: `program == 'survey'` exposures with the standard 90 s exposure time, with quality cuts dropping rows carrying unphysical pipeline values (missing or non-positive $t_{\rm eff}$; FWHM $\leq 0.4''$ or FWHM $\geq 4''$; sky excess ≤ -0.9; corrupt or epoch-zero timestamps). These cuts remove sensor and pipeline artifacts, not real observations. The clean sample contains 74,323 exposures (g: 20,128; r: 18,032; i: 16,971; z: 15,629; Y: 3,563) spread over 615 scoreable nights. Exposures are grouped into local observing nights by shifting the UTC timestamps back 12 hours, so that a single dark period is never split at midnight.

3 The effective exposure time

3.1 The formula and where it comes from

The metric we adopt is the effective exposure time of Neilsen et al. (2019):

$$t_{\rm eff} \;=\; \eta^2 \left(\frac{\rm FWHM_{fid}}{\rm FWHM}\right)^2 \left(\frac{b_{\rm dark}}{b}\right), \tag{1}$$

where η is the atmospheric transmission (unity in photometric conditions), FWHM is the delivered image blur, b is the sky background surface brightness, and the "fid"/"dark" subscripts denote fiducial benchmark conditions. Each factor, and in particular each *exponent*, follows from the standard signal-to-noise argument for a faint point source on a bright sky. It is worth walking through, because the exponents are exactly what make $t_{\rm eff}$ a good reward.

Consider a point source of flux F observed for clock time τ. The detected signal is $S \propto \eta F \tau$: photons are collected linearly in time, and clouds remove a fraction $(1-\eta)$ of them. The source's light lands on the detector in a seeing disk of diameter $\sim$ FWHM, i.e. spread over an area of $n_{\rm pix} \propto {\rm FWHM}^2$ pixels. In the background-limited regime (the regime of every faint-object survey), the noise is dominated by Poisson fluctuations of the sky counts accumulated under the source footprint: $N \propto \sqrt{b\, n_{\rm pix}\, \tau} \propto {\rm FWHM}\sqrt{b\,\tau}$. Therefore

$$\mathrm{SNR} \;=\; \frac{S}{N} \;\propto\; \frac{\eta F \tau}{\mathrm{FWHM}\sqrt{b\,\tau}} \qquad\Longrightarrow\qquad \mathrm{SNR}^2 \;\propto\; \frac{\eta^2}{\mathrm{FWHM}^2\, b}\,\tau. \tag{2}$$

Squared SNR—the statistically meaningful "information content" of the exposure—accrues *linearly in time*, with a proportionality constant set entirely by conditions. Now ask: how long an exposure $\tau_{\rm eff}$ in fiducial conditions ($\eta = 1$, $\mathrm{FWHM_{fid}}$, $b_{\rm dark}$) would reach the same SNR^2 as our actual exposure of length τ? Setting the two expressions equal and solving gives $\tau_{\rm eff} = t_{\rm eff} \cdot \tau$ with $t_{\rm eff}$ exactly as in Eq. (1). The exponents are now transparent:

- **Transmission enters squared** (η^2): the signal scales with η, and SNR^2 scales with signal squared over noise—the sky noise does not care about the clouds dimming the source. Half a magnitude of cloud ($\eta = 10^{-0.4\times 0.5}$) already costs 60% of the exposure's value.
- **Blur enters squared**: a point source's flux is spread over an area $\propto \mathrm{FWHM}^2$, so the sky noise under the source grows linearly with FWHM, and SNR^2 falls as FWHM^{-2}. Doubling the seeing quarters the worth of the exposure.
- **Sky brightness enters linearly**: it appears only in the noise variance, which is proportional to the background count rate.

3.2 Why this is the natural scheduler reward

The derivation gives $t_{\rm eff}$ a precise operational meaning: *an exposure of length τ taken in the given conditions detects point sources exactly as well as an exposure of length $t_{\rm eff}\,\tau$ taken in fiducial conditions.* $t_{\rm eff}$ is a multiplier on time itself. That interpretation is what elevates it from a quality statistic to a reward: the quantity

$$\sum_{\rm exposures} t_{\rm eff} \cdot \tau \tag{3}$$

is the night's *effective open-shutter time*—science-grade seconds in a common currency—and a scheduler that maximizes it is maximizing precisely the resource a survey exists to collect. Rewards

in different bands, sky conditions, and moon phases all become commensurable seconds. This is the quantity Terranova et al. (2023) used as the per-step DQN reward, and the quantity we adopt.

DES itself uses $t_{\rm eff}$ operationally: exposures falling below per-band thresholds ($t_{\rm eff} < 0.2$ in g and Y; $t_{\rm eff} < 0.3$ in r, i, z) do not count toward survey completion and are retaken (Neilsen et al., 2019). These thresholds give us a natural per-exposure pass/fail criterion that we reuse in Section 6.

4 Empirical calibration and validation

Eq. (1) is a physical form, not yet a computable function of the CSV columns: we must identify what `qc_cloud` and `qc_sky` actually measure and what fiducial values DES used. Rather than trusting constants quoted in papers, we fit everything on the data and let the DES pipeline's own `qc_teff` arbitrate. This section is the heart of the phase-one work. All comparisons below use log-space correlation, since $t_{\rm eff}$ spans more than two decades and multiplicative errors are the relevant ones.

4.1 Step 1: the blur × cloud core identifies `qc_cloud`

We first test the hypothesis that `qc_cloud` is a cloud extinction in magnitudes, so that $\eta^2 = 10^{-0.8\,c}$. Using only the blur and cloud factors,

$$t_{\rm eff}^{\rm core} = 10^{-0.8\,c} \left(\frac{\rm FWHM_{fid}}{\rm FWHM} \right)^2, \tag{4}$$

already reproduces the pipeline $t_{\rm eff}$ with a log-correlation of 0.93 across the full clean sample. This simultaneously confirms the magnitude interpretation of `qc_cloud` and shows the multiplicative structure of Eq. (1) is the right skeleton; the residual scatter must be carried by the sky term.

4.2 Step 2: per-band fiducial seeing—and a consistency check against physics

The fiducial $\rm FWHM_{fid}$ is band-dependent (bluer light is blurred more by the atmosphere), and its values are a convention baked into the DES pipeline. We fit them per band by requiring our reconstruction to match `qc_teff` in the median. The results are listed in Table 1.

Table 1: Per-band calibration fit on the clean 90 s survey sample. Fiducial FWHM is the seeing at which the blur term equals unity; α_b is the exponent of the empirical sky term $(1 + \Delta b)^{-\alpha_b}$; the last row gives the DES keep-or-retake threshold on $t_{\rm eff}$ (Neilsen et al., 2019).

	g	r	i	z	Y
Fiducial FWHM (arcsec)	1.119	0.991	0.932	0.871	0.904
Sky exponent α_b	0.589	0.487	0.254	0.213	0.881
$t_{\rm eff}$ threshold	0.2	0.3	0.3	0.3	0.2

These fitted numbers carry an independent physics check. Kolmogorov turbulence theory predicts that seeing scales with wavelength as $\rm FWHM \propto \lambda^{-1/5}$; if our fitted fiducials are physically meaningful they should trace this law across the five DECam bands ($\lambda_{\rm eff}$ = 475, 635, 775, 925, 1000 nm for g, r, i, z, Y). They do (Figure 1): four of the five bands fall on the $\lambda^{-0.2}$ line, with Y the mild outlier—plausibly because Y is used mostly in bright time and contributes the smallest

sample. Fitting constants on data and *rediscovering* an atmospheric scaling law we did not impose is the kind of consistency that makes us trust the calibration.

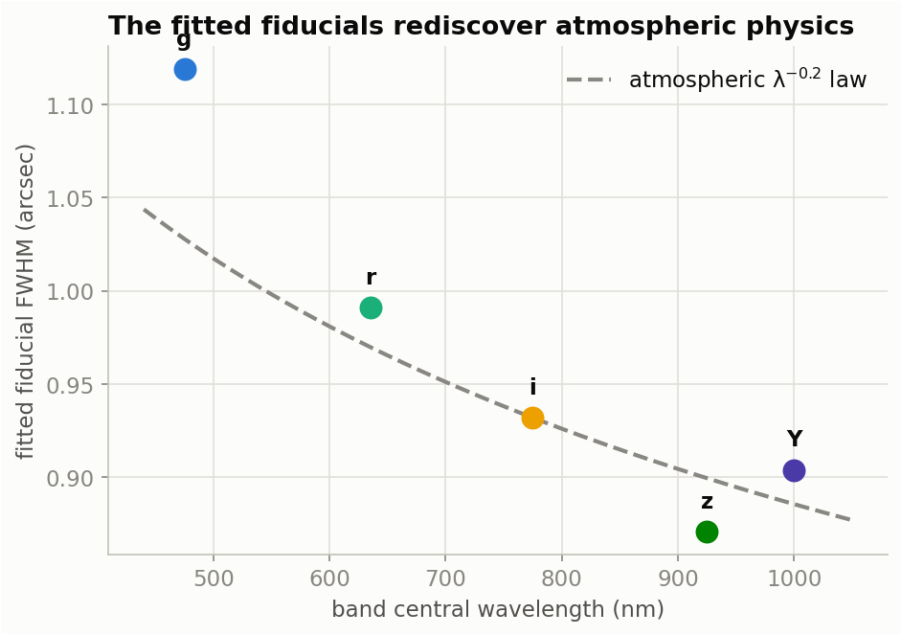


Figure 1: Fiducials rediscover physics: the per-band fiducial FWHM, fit purely to match the DES pipeline $t_{\rm eff}$, follows the atmospheric $\lambda^{-0.2}$ law.

4.3 Step 3: an empirical sky term

The released `qc_sky` column is not a surface brightness in physical units; it behaves as a *fractional* excess Δb over nominal dark sky. The ideal sky factor $b_{\rm dark}/b$ would then be $(1+\Delta b)^{-1}$, but the pipeline's $t_{\rm eff}$ is evidently computed from a sky model that responds differently per band (the sky spectrum, the detector response, and the definition of "nominal" all vary with filter). We therefore adopt the empirical form

$$\text{sky term} = (1+\Delta b)^{-\alpha_b}, \tag{5}$$

with one exponent per band, fit against `qc_teff` with the blur and cloud terms in place. The fitted α_b (Table 1) are largest in g and Y and smallest in z—consistent with moonlight contaminating the blue bands most strongly while z is dominated by airglow, though we treat α_b as calibration constants rather than over-interpreting them.

4.4 Step 4: validation against the DES pipeline

Figure 2 shows the full calibrated reconstruction against the pipeline's `qc_teff` across all 74,323 clean exposures, and Figure 3 shows the three factors individually as multipliers on exposure time. The headline validation numbers:

The calibration buys a substantial improvement over plugging in nominal constants: the median relative error drops from 23.9% to 14.8%, and the log-correlation reaches 0.952. The remaining scatter is consistent with the pieces we cannot see from the released columns (the pipeline's full sky model and any per-CCD effects) and is small compared to the dynamic range of $t_{\rm eff}$ itself, which spans from clouded-out ($t_{\rm eff} \approx 0.02$) to better-than-fiducial ($t_{\rm eff} > 1.5$) conditions.

Table 2: Validation of the reconstructed t_{eff} against the DES pipeline value, over 74,323 clean 90 s survey exposures.

Model	log-correlation	median |relative error|
Blur × cloud core only	0.933	—
Naive paper constants (no calibration)	—	23.9%
Full calibrated model	**0.952**	**14.8%**

The entire calibration is packaged in the `skai_metrics` toolkit (`teff.py`, `blur.py`, `schedule.py`, `data.py`) together with an automated test suite that re-derives these checks from the raw CSV; all **4 of 4 validation tests pass**, so the numbers in this section are reproducible with one command.

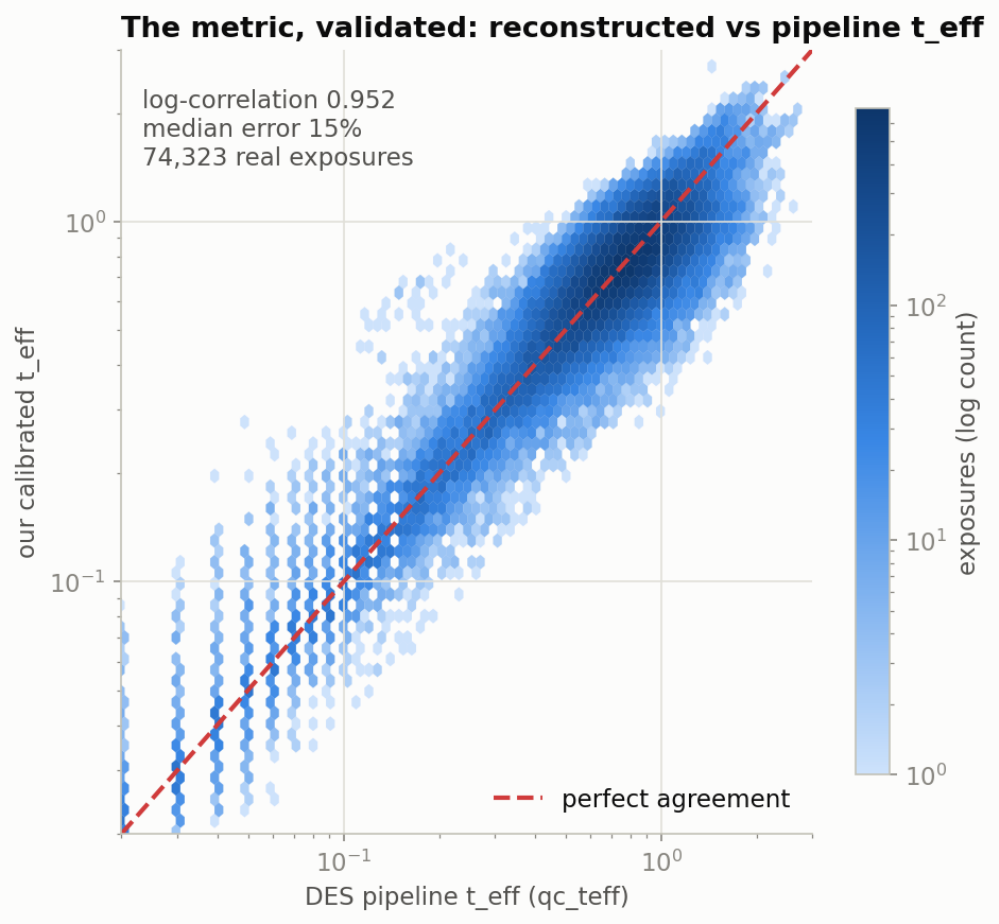


Figure 2: Reconstructed vs. pipeline t_{eff}: our calibrated metric against DES's own quality pipeline across 74,323 exposures—log-correlation 0.952.

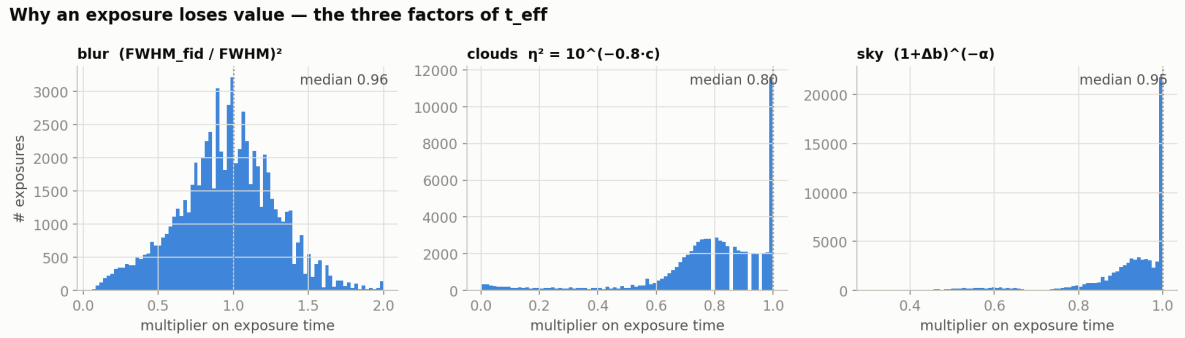


Figure 3: The three factors of $t_{\rm eff}$—blur, cloud, and sky—each acting as a multiplier on exposure time.

5 The blur metric, decomposed: what the atmosphere did vs. what the scheduler chose

5.1 The decomposition

The blur term is the largest and most structured factor of $t_{\rm eff}$, and it mixes two things a reward must keep apart. Delivered image quality decomposes as

$$\mathrm{FWHM}_{\rm delivered} \;\simeq\; \mathrm{FWHM}_{\rm zenith} \times X^{0.6} \times \left(\frac{\lambda}{\lambda_{\rm ref}}\right)^{-0.2}, \tag{6}$$

where $\mathrm{FWHM}_{\rm zenith}$ is the seeing the atmosphere is delivering straight up—*tonight's weather, which the scheduler cannot control*—and $X^{0.6}$ is the airmass penalty, which the scheduler pays *by choice* every time it points away from zenith. (The 0.6 exponent is Kolmogorov turbulence again: the turbulence integral grows with path length $\propto X$, and seeing scales as its 3/5 power. The $\lambda^{-0.2}$ factor is the same chromatic law of Figure 1; band choice is usually dictated by survey needs, so we treat it as neither weather nor scheduling merit.) The penalty is steep in reward terms: since the blur term of $t_{\rm eff}$ goes as FWHM^{-2}, pointing at $X = 1.5$ costs a factor $X^{-1.2} \approx 0.61$—the scheduler throws away 39% of the blur term relative to zenith.

A reward that punishes the agent for the atmosphere teaches it nothing; a reward blind to airmass forgives its worst habit. The `blur` module therefore reduces every exposure to a common axis (zenith seeing in the i band, via Eq. (6)) and reports the airmass factor separately. Figure 4 shows the delivered blur distributions per band (redder bands are sharper, as the $\lambda^{-0.2}$ law demands; the medians run from $1.14''$ in g to $0.92''$ in z), and Figure 5 shows the strong seasonality of zenith seeing at CTIO—the Chilean summer delivers visibly sharper images, which will matter when comparing schedules from different seasons. Over the whole survey, the median zenith-reduced i-band seeing is $0.87''$ against a median delivered i-equivalent of $0.97''$, and the median exposure loses 19.7% of its blur-term $t_{\rm eff}$ to the pointing (airmass) decision alone.

5.2 The headline finding: Simpson's paradox in the scheduler logs

Eq. (6) makes a clean prediction: regressing $\log \mathrm{FWHM}$ on $\log X$ should return a slope of $+0.6$. On the pooled DES sample it returns

$$\text{pooled fit:}\quad \mathrm{FWHM} \propto X^{-0.23},$$

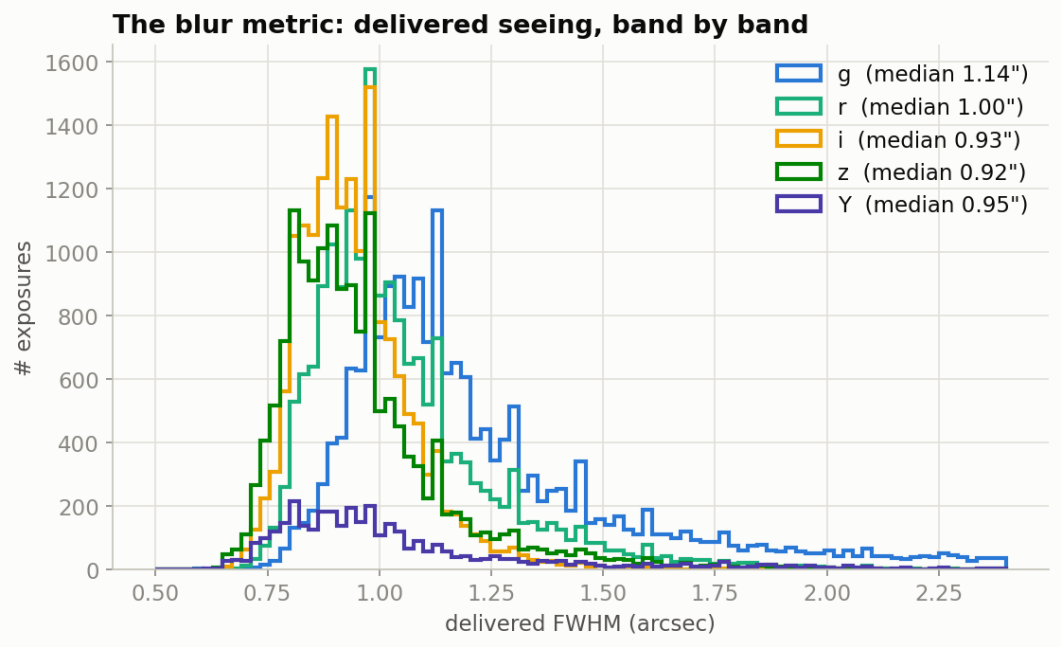


Figure 4: Delivered seeing by band: blur distributions in each DECam filter. Redder bands are sharper, following the $\lambda^{-0.2}$ law.

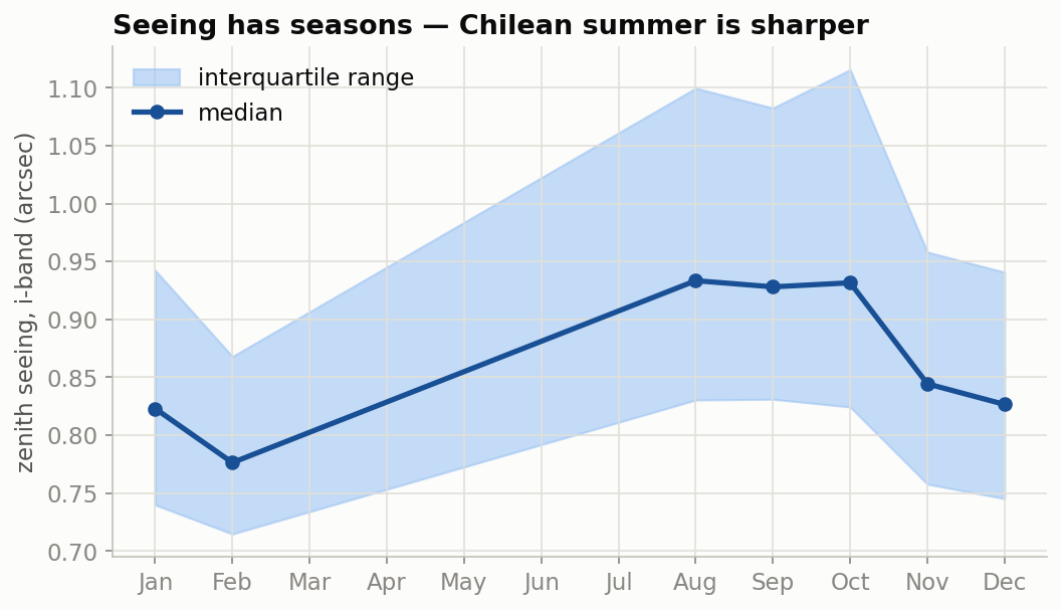


Figure 5: Seeing seasonality: monthly zenith seeing at CTIO. The Chilean summer delivers sharper images.

i.e. image blur apparently *improves* as the telescope points farther from zenith. Taken at face value this contradicts atmospheric physics. The resolution is that the data are not an experiment—they are the log of a competent scheduler acting on the conditions. DES pointed at high airmass *preferentially when the seeing was good*: across exposures, the correlation between zenith seeing and chosen airmass is -0.36. In good weather the scheduler could afford to chase fields far from zenith (and did, to serve footprint and cadence goals); in bad weather it hugged the zenith to salvage image quality. Conditioning away the nightly weather recovers the physics: a within-night fixed-effects fit (each night keeps its own intercept, so only *within-night* airmass variation identifies the slope) yields an exponent of $+0.23$, restoring the physical sign and moving substantially toward the theoretical $+0.6$ (attenuated by intra-night seeing drifts and measurement noise). Figure 6 shows both fits; Figure 7 shows the atmosphere/scheduler split and the $t_{\rm eff}$ fraction lost to pointing, and Figure 8 maps delivered seeing across the DES footprint.

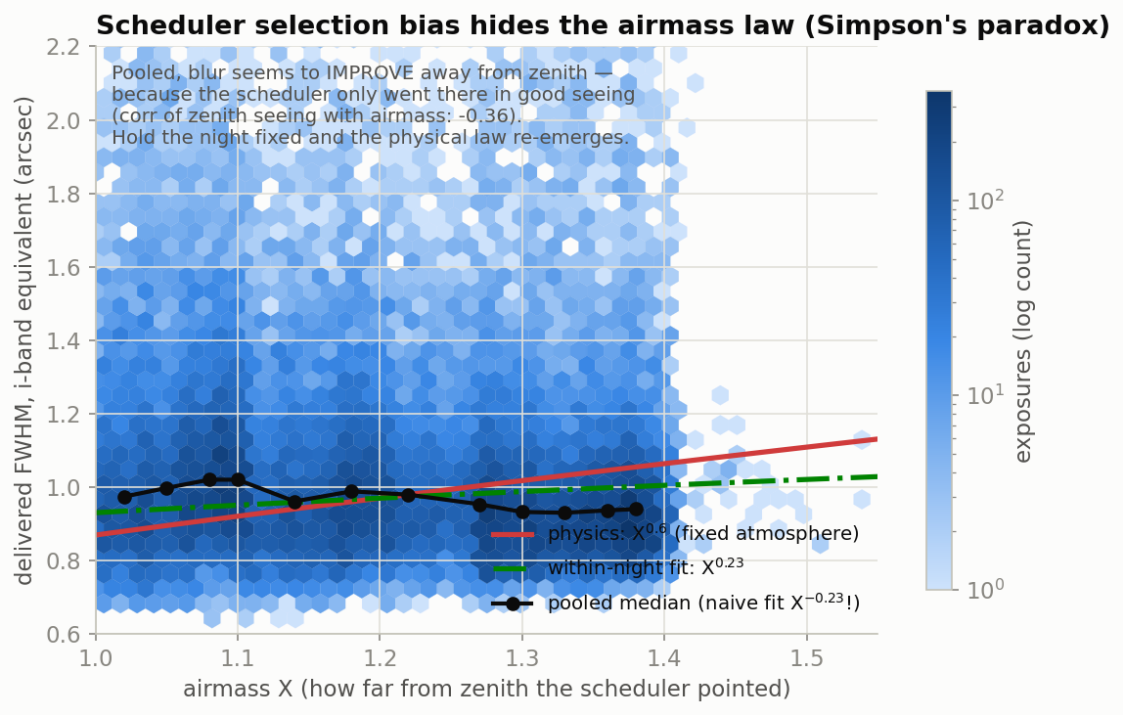


Figure 6: Scheduler selection bias as Simpson's paradox. Pooled over all exposures, blur seems to improve with airmass (slope -0.23), because the scheduler pointed low only in good seeing (corr(zenith seeing, airmass) $= -0.36$). Holding the night fixed recovers the physical $X^{0.6}$ trend (within-night slope $+0.23$).

This is textbook Simpson's paradox—a trend that reverses once a confounder (the night's weather) is conditioned on—but we want to stress what it means for our research program. The confounder here is not a nuisance of nature; *it is the logging policy itself*. Any naive evaluation of a scheduling policy against these logs inherits the bias: a policy that points at high airmass would look spuriously good, because in the data high-airmass exposures happen to be good-seeing exposures. This is exactly the off-policy evaluation problem, encountered empirically in our own data before we have trained a single agent. It sharpens the case for two design decisions: (i) rewards and evaluation metrics must be built from the *decomposed* quantities of Eq. (6)—credit the scheduler for its airmass choices given the atmosphere it was dealt, never for the atmosphere

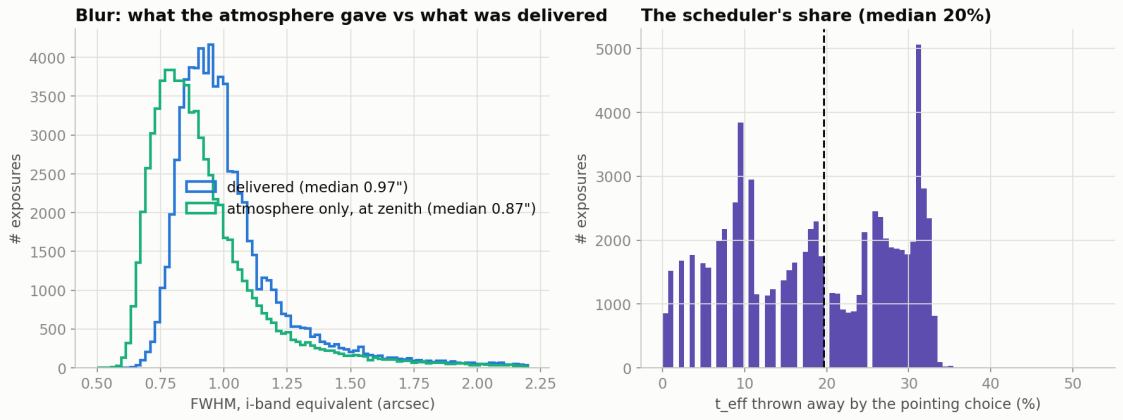


Figure 7: Atmosphere vs. scheduler: zenith seeing (uncontrollable) against delivered blur, and the fraction of blur-term t_{eff} lost to the pointing decision (median 19.7%).

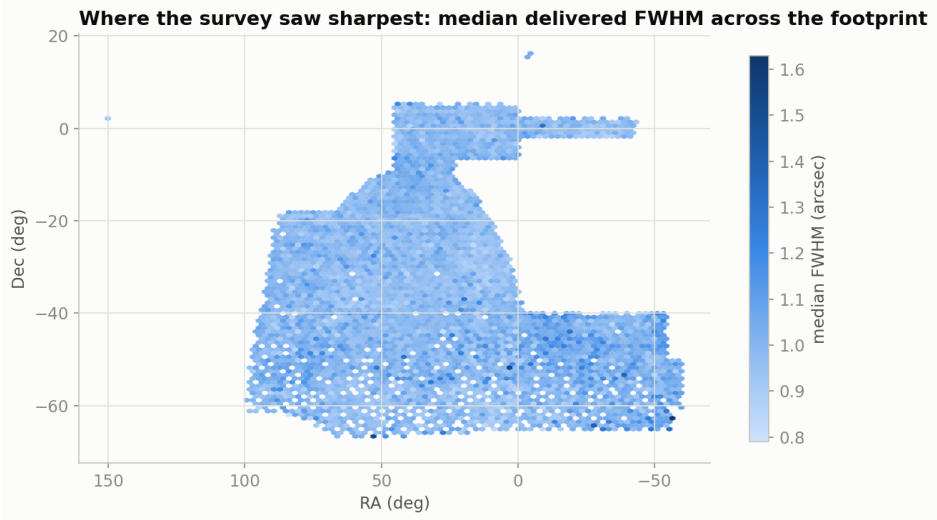


Figure 8: Median delivered FWHM across the DES footprint.

itself; and (ii) on the learning side, conservative offline-RL methods such as CQL (Kumar et al., 2020), which explicitly distrust state–action regions the logging policy never visited, are not optional safety trimmings but responses to a bias we can now point to in the data.

6 From exposures to schedules: per-night metrics

6.1 Metrics

$t_{\rm eff}$ scores one exposure; a schedule is a night of sequential decisions, and its score must also account for the time the telescope wasted *between* exposures. The bridge is Eq. (3): the night's effective science seconds are $\sum t_{\rm eff}\,\tau$, and our headline schedule metric is the fraction of the wall-clock night converted into fiducial-condition science time,

$$\text{effective_efficiency} \;=\; \frac{\sum_i t_{\text{eff},i}\,\tau_i}{\text{night span}}, \tag{7}$$

where the night span runs from the first shutter-open to the last shutter-close. This single number is degraded by *either* of the two independent ways a night is lost—sitting idle (dead time between exposures) or observing through poor conditions—and Figure 10 separates the two by plotting open-shutter efficiency against mean $t_{\rm eff}$.

Because a single number hides *why* a night scored what it did, we decompose it into three sub-scores, each already on a $[0, 1]$ scale:

- **yield** — the effective efficiency of Eq. (7) (capped at 1): everything combined.
- **quality** — the fraction of exposures passing the DES per-band keep-or-retake thresholds (Table 1; Figure 12 shows the pass fractions per band across the survey): would this night's data actually count?
- **pointing** — the mean fraction of the blur term retained given the airmasses the scheduler chose, $\overline{X^{-1.2}}$ (1.0 = everything at zenith): the sub-score that isolates the scheduler's own contribution, following Section 5. Clouds and bad seeing are not its fault; airmass is a choice.

These combine into a composite 0–100 score as a weighted average with weights 0.5/0.25/0.25 (yield/quality/pointing). We state plainly: *these weights are a v0.1 proposal*, chosen to be transparent and decomposable rather than final, and reviewing them with our mentors is the first item of Section 7. The value of the construction is that any reweighting is a one-line change and every score remains auditable into its three named parts.

6.2 Results on 615 real DES nights

Applying the metrics to the clean survey sample scores 615 nights; Figure 9 shows every night with a 60-day rolling median. The median composite score is 64.1, with clear seasonal structure inherited from the seeing seasonality of Figure 5. Table 3 lists the five best and five worst nights, and Figure 11 follows $t_{\rm eff}$ exposure-by-exposure through the extreme nights.

The extremes are physically sensible, which is itself a validation. The best night, 2019-01-08 (Chilean summer, near the survey's end), has a mean $t_{\rm eff}$ of 1.69—conditions *better than fiducial*, so every open-shutter second was worth 1.69 fiducial seconds, and the effective efficiency exceeds unity (1.19): the night produced more fiducial-condition science time than its wall-clock span. Every exposure passed threshold, and the median airmass of 1.06 shows the scheduler was also

Table 3: The five best and five worst DES nights under the composite score. "Eff. eff." is the effective efficiency of Eq. (7); "pass" is the fraction of exposures meeting the DES $t_{\rm eff}$ thresholds; $X_{\rm med}$ is the median airmass.

Night	Score	$N_{\rm exp}$	Mean $t_{\rm eff}$	Eff. eff.	Pass	$X_{\rm med}$
Best						
2019-01-08	98.3	34	1.693	1.190	1.000	1.06
2015-09-01	97.8	27	1.532	1.074	1.000	1.08
2018-11-26	96.2	218	1.659	1.128	1.000	1.15
2018-02-21	96.0	96	1.430	1.054	0.990	1.16
2017-12-21	93.8	63	1.317	0.992	1.000	1.28
Worst						
2018-09-11	19.2	24	0.021	0.010	0.000	1.31
2016-09-14	20.8	27	0.068	0.013	0.000	1.19
2014-01-17	21.9	26	0.161	0.089	0.000	1.37
2014-10-14	22.9	77	0.080	0.041	0.013	1.19
2015-09-15	23.3	57	0.054	0.019	0.000	1.09

pointing well. The worst night, 2018-09-11, was simply clouded out: mean $t_{\rm eff}$ of 0.02 means the shutter was open but essentially nothing came through—an effective efficiency of 0.01, with not one exposure passing threshold. Note that its open-shutter efficiency (0.50) was unremarkable: the telescope was *operating*, just pointlessly. Only the effective efficiency, not any shutter-time bookkeeping, captures that distinction—the efficiency plane of Figure 10 makes the two failure axes explicit.

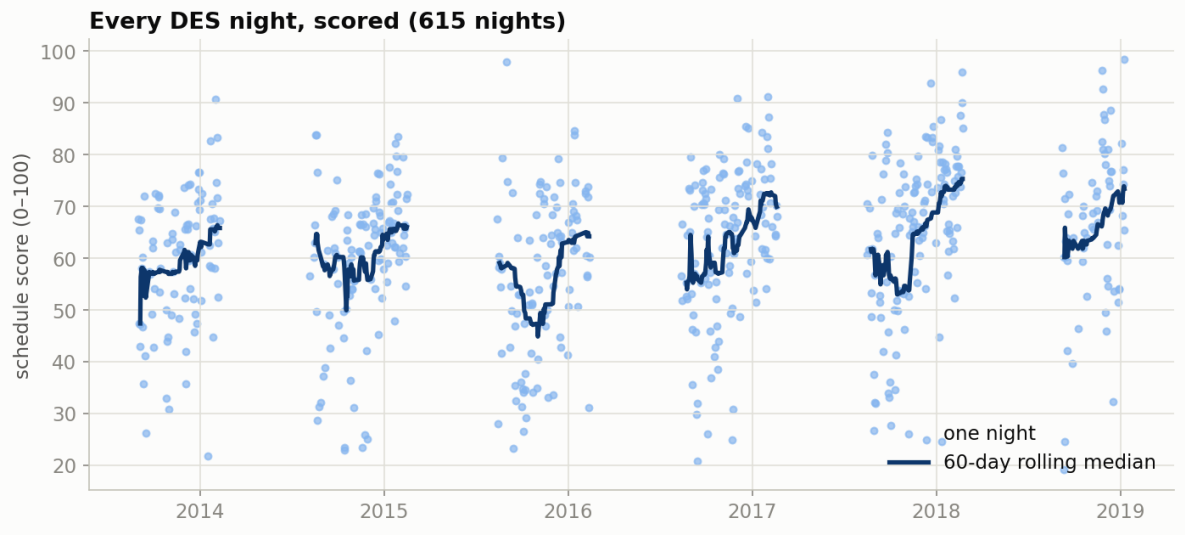


Figure 9: Every DES night, scored: the composite schedule score for each of the 615 nights with a 60-day rolling median.

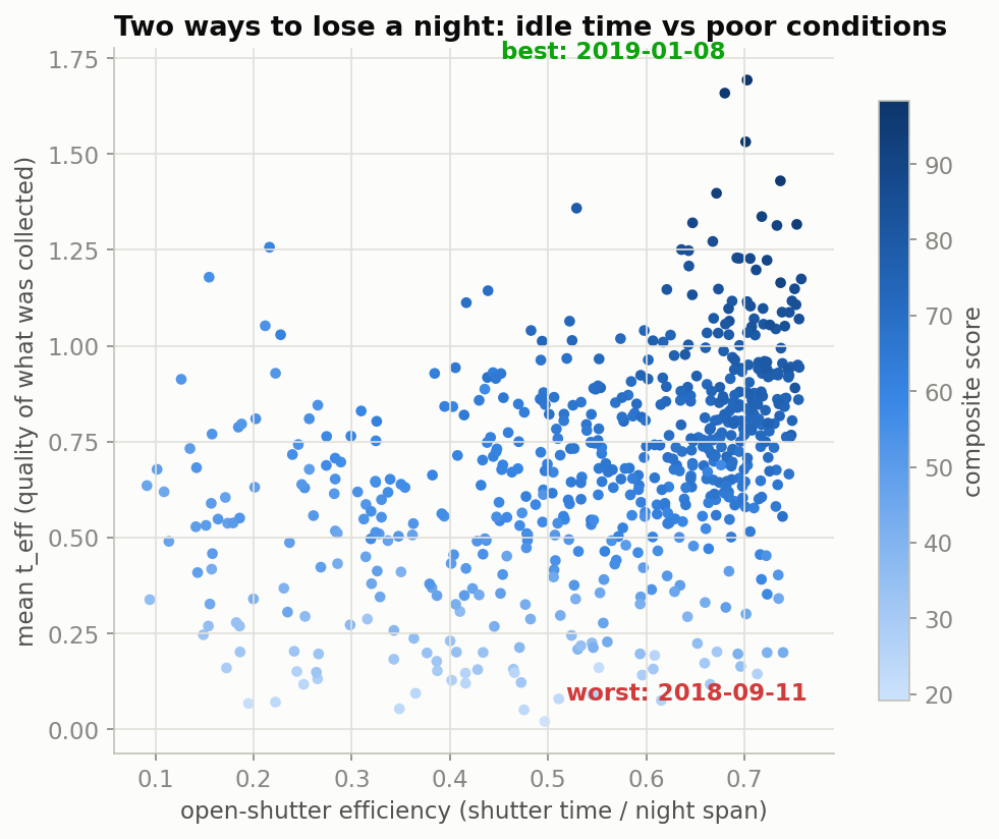


Figure 10: The efficiency plane: idle time vs. poor conditions are two independent ways a night is lost.

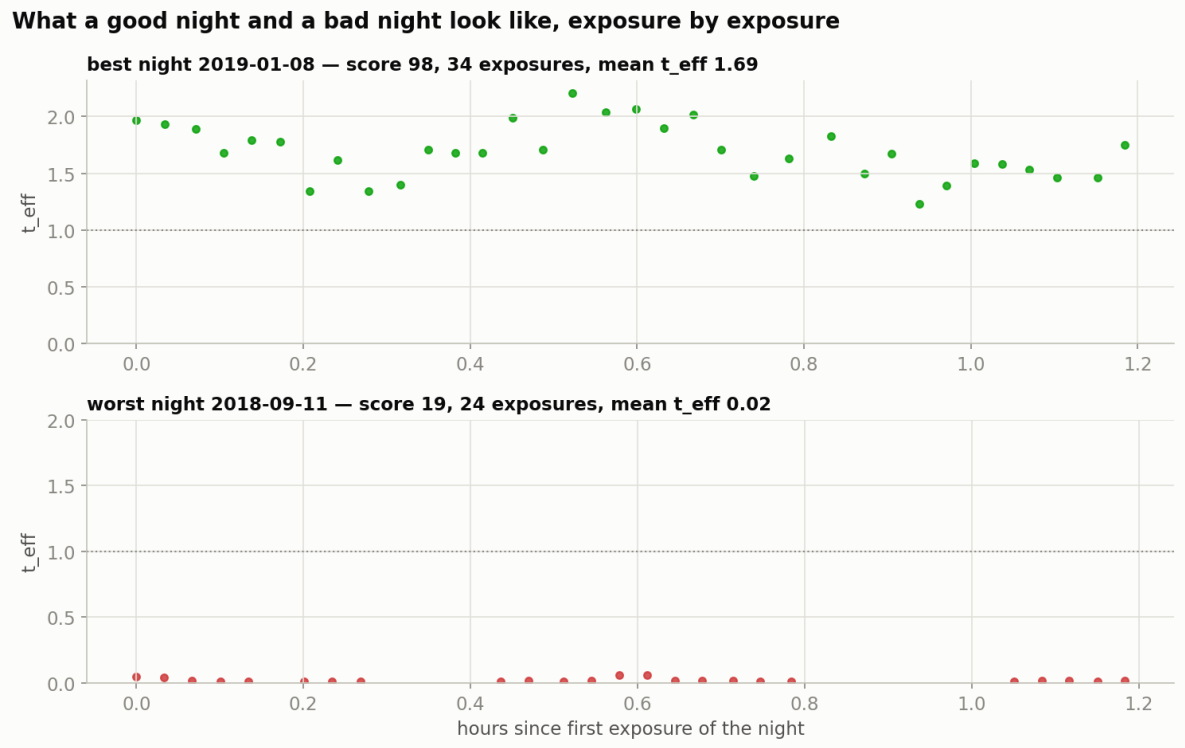


Figure 11: Exposure-by-exposure t_{eff} through the highest- and lowest-scoring nights.

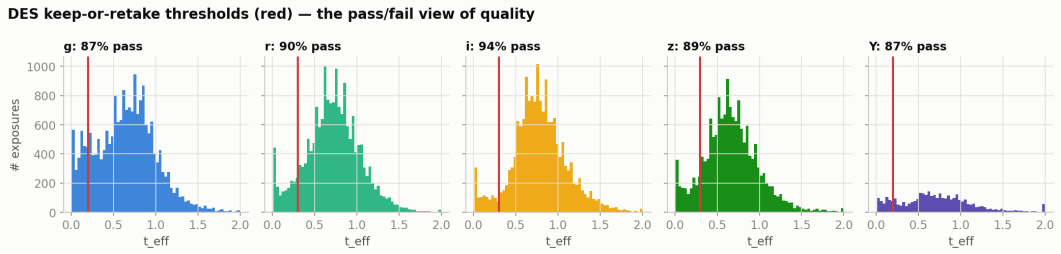


Figure 12: Keep-or-retake: fraction of exposures meeting the DES per-band survey thresholds on t_{eff}.

7 Discussion and next steps

What this phase establishes: a per-exposure reward ($t_{\rm eff}$) with a first-principles derivation, calibrated per band and validated at log-correlation 0.952 against the DES pipeline; a decomposition of its dominant term that separates weather from scheduling decisions; schedule-level metrics with a transparent composite score, exercised on 615 real nights; and a packaged, tested toolkit (`skai_metrics`) the group can build on. It also surfaced a finding we did not go looking for—the airmass/seeing selection bias of Section 5.2—which turns an abstract concern about off-policy evaluation into a measured number in our own data.

The immediate next steps, roughly in order:

1. **Review the composite weights with mentors.** The 0.5/0.25/0.25 split of yield/quality/pointing is a deliberate v0.1 stake in the ground. The sub-scores are the durable part; the weights should reflect the group's priorities and may end up program-dependent.

2. **Score the group's prototype AI scheduler.** The metrics were built to compare schedulers, not just nights. Running the toolkit on the prototype's simulated output against matched DES nights is the first real test—with the Section 5.2 caveat front and center whenever the comparison leans on logged conditions.

3. **Per-program weighting.** The present scores treat all effective seconds equally, which is right for the wide survey but wrong across programs: a supernova-field visit has cadence value and a GW follow-up exposure has time criticality that raw $t_{\rm eff} \cdot \tau$ does not see. A per-program utility weight on Eq. (3) is the natural extension.

4. **Slew and dead-time modeling.** Effective efficiency penalizes dead time without attributing it. Modeling slew time as a function of angular distance between pointings would let us split dead time into "physically necessary" and "wasted," giving the scheduler a sharper signal about sequence ordering.

5. **Toward reward shaping for CQL.** The end state is a reward for conservative offline RL (Kumar et al., 2020): per-step $t_{\rm eff} \cdot \tau$ as the base, with the pointing decomposition guarding against crediting weather, and the per-program weights encoding scientific priorities. The evaluation metrics of Section 6 then serve as the held-out judge of whatever policy the training produces.

The guiding principle throughout is the one that opened this note: the reward is the foundation. We would rather spend a phase validating it against 74,323 real exposures than discover, several training runs from now, that our agent has been diligently optimizing an artifact.

References

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